

README

Overview

The code in this replication package constructs the analysis file from the 11 data sources (the *2018 Multiple Indicator Cluster Survey (MICS)*, the *10% sample of the Lesotho Population and Housing Census of 2006*, the *Lesotho Demographic and Health Surveys of 2004-05, 2009-10, and 2014 (DHS)*, the *Lesotho Southern and Eastern Africa Consortium for Monitoring Educational Quality 2000 and 2007 Surveys (SACMEQ)*, the *Lesotho 1965 School enrolment data*, the *10% sample of the South Africa Census of 2001*, the *8.5% sample of the South Africa Census of 2011*, and the *5.8% sample of the South Africa Community Survey of 2016*) using Stata. There is also supplementary data from 8 different studies (cited below) on effect sizes for years of schooling and GDP per capita (current US\$) from the World Bank. The “AEJ_Datasets_3 Jun 2025” dofile prepares the datasets used in the analysis, and the dofiles “AEJ_Figures_3 Jun 2025” and “AEJ_Tables_3 Jun 2025” generate all 18 figures and two tables in the paper, and all the figures and tables in the Appendix. Dofile “_masterscript.do” runs the entire analysis. The replicator should expect the code to run for approximately 2 minutes on a MacBook Pro (2024 M4).

Data Availability and Provenance Statements

Statement about Rights

- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☒ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permissions are documented in the [LICENSE.txt](#) file.

(Optional, but recommended) License for Data

- ☒ See LICENSE.txt for details.

Summary of Availability

- ☒ All data are publicly available.

Details on each Data Source

Data.Name	Data.Files	Location	Provided	Citation
“Multiple Indicator Cluster Survey (MICS) 2018”	mics6hl.dta; mics6fs.dta; micsbh.dta; mics2018_wm_mn.dta	Raw_data/MICS	TRUE	Bureau of Statistics (2019)

<i>“Lesotho Population and Housing Census of 2006”</i>	census lesotho.dta	Raw_data /IPUMS/Lesotho	TRUE	IPUMS (2019)
<i>“Lesotho Demographic and Health Surveys of 2004-05, 2009-10, and 2014 (DHS)”</i>	dhsbh.dta; dhshiv.dta; dhskids.dta; dhsmen.dta; dhspooledPR.dta; dhswomen.dta	Raw_data/DHS	TRUE	ICF. 2004-2017.
<i>Lesotho Southern and Eastern Africa Consortium for Monitoring Educational Quality 2000 and 2007 Surveys (SACMEQ)</i>	sacmeqiii_ii.dta	Raw_data/SACMEQ	TRUE	Southern Africa Consortium on Monitoring Educational Quality (SACMEQ)
<i>Lesotho 1965 School enrolment data</i>	Enrolment data.xlsx	Raw_data/BASUTO LAND	TRUE	Basutoland (1965)
<i>South Africa Census of 2001, South Africa Census of 2011</i>	census_south afr_all.dta	Raw_data/IPUMS/ South_Africa	TRUE	IPUMS (2019)
<i>South Africa Community Survey of 2016</i>	census_south afr_all.dta	Raw_data/IPUMS/ South_Africa	TRUE	IPUMS (2019)
<i>Data on effect sizes for years of schooling from selected existing studies, and</i>	Effect size data.xlsx	Raw_data/ EFFECT SIZES	TRUE	Black et al (2011); Fredriksson and Ockert (2014); Arnold and Depew

*GDP per
capita of
selected
countries*

(2018);
McCrary and
Royer
(2011);
Fredriksson,
Huttunen
and Ockert
(2022); Guo
et al (2023);
Kawaguchi
(2011);
Valdes and
Requena
(2024);
World Bank
(2024)

Public use data sourced from elsewhere and provided

Data on 1965 school enrollment was derived from the Annual Report of the Basutoland Director of Education for the year 1965 (Basutoland, 1965). A copy of the data is provided as part of this archive. The report is in the archives of the National University of Lesotho.

Datafile: Raw_data/Enrolment data.xlsx

Data on effect sizes for years of schooling in different countries was sourced from selected existing studies (Black et al., 2011; Fredriksson and Ockert, 2014; Arnold and Depe, 2018; McCrary and Royer, 2011; Fredriksson, Huttunen and Ockert, 2022; Guo et al., 2023; Kawaguchi, 2011; and Valdes and Requena, 2024). Data on GDP per capita (current US\$) was retrieved from the World Bank (2024).

Datafile: Raw_data/Effect size data.xlsx

Public use data with required registration and provided extract

The paper uses the Lesotho 2006 Census data, the South Africa Census of 2001, the South Africa Census of 2011, and the South Africa Community Survey of 2016 from IPUMS (Minnesota Population Center, 2019). IPUMS does not allow for redistribution, except for the purpose of replication archives. We merged the South Africa Census of 2001, the South Africa Census of 2011, and the South Africa Community Survey of 2016. To obtain data, one needs to first register and then select the respective datasets.

Datafiles: Raw_data/IPUMS/Lesotho/census_lesotho.dta;
Raw_data/IPUMS/South_Africa/census_south afr_all.dta

The paper uses the Lesotho Multiple Indicator Cluster Survey (MICS) of 2018 (Bureau of Statistics, 2019). The Bureau of Statistics and UNICEF do not allow for redistribution, except for the purpose of replication archives. Data can also be directly downloaded from <https://mics.unicef.org/surveys>. To obtain data, one needs to first register.

Datafile: Raw_data/MICS/mics6hl.dta; Raw_data/MICS/mics6fs.dta;
Raw_data/MICS/micsbh.dta; Raw_data/MICS/mics2018_wm_mn.dta

The paper also uses the Lesotho Demographic and Health Surveys of 2004-05, 2009-10, and 2014 (DHS) (ICF. 2004-2017). The ICF does not allow for redistribution, except for the purpose of replication archives. Data can also be directly downloaded from <https://dhsprogram.com/Data/>. To obtain data, one needs to first register.

Datafile: Raw_data/DHS/dhsbh.dta; Raw_data/DHS/dhshiv.dta; Raw_data/DHS/dhskids.dta;
Raw_data/DHS/dhsmen.dta; Raw_data/DHS/dhspooledPR.dta;
Raw_data/DHS/dhswomen.dta

The Lesotho Southern Africa Consortium on Monitoring Educational Quality (SACMEQ) data, SACMEQ II and SACMEQ III (Southern Africa Consortium on Monitoring Educational Quality (SACMEQ 2005-2013). The SACMEQ does not allow for redistribution, except for the purpose of replication archives. Data can also be directly downloaded from <http://www.seacmeq.org/>. To obtain data, one needs to first register.

Datafile: Raw_data/SACMEQ/sacmeqiii_ii.dta

Preliminary code during the editorial process

Code for data cleaning and analysis is provided as part of the replication package. It has been uploaded to the AEA Data and Code Repository.

Computational requirements

Software Requirements

- ☒ The replication package contains one or more programs to install all dependencies and set up the necessary directory structure. [HIGHLY RECOMMENDED]
- Stata (code was last run with version 17.0)
 - outreg2 (version 2.3.2, 17-08-2014)

- outreg (version 4.32, 18-09-2015)
- rdrobust (version 9.0.5, 2022-09-29)
- blind schemes (Version 1.3.0)
- coefplot (version 1.8.6 22-02-2023)
- rddensity (version 2.3 2021-02-28)
- lpdensity (version 2.5 2024-10-06)
- statplot (version 1.2.7 5 May 2011)
- addplot (version 1.0.1 25-09-2014)
- schemepack (version 1.4, 09-09-2022)

Memory, Runtime, Storage Requirements

Summary

The approximate time needed to reproduce the analyses on a standard (2025) desktop machine is

- ☒ <3 minutes

Approximate storage space needed:

- ☒ 2 GB - 25 GB

Description of programs/code

- The program “Dofiles/AEJ_Datasets_3 Jun 2025” prepares the datasets used in the analysis.
- The programs “Dofiles/AEJ_Figures_3 Jun 2025” and “Dofiles/AEJ_Tables_3 Jun 2025” generate all 18 figures and two tables in the paper and all the figures and tables in the Appendix.
- Program “Dofiles/_masterscript.do” runs the entire analysis (i.e., all other programs).

Instructions to Replicators

- Programs “Dofiles/AEJ_Figures_3 Jun 2025” and “Dofiles/AEJ_Tables_3 Jun 2025” will create all output directories and install needed ado packages (uncomment, i.e., remove the * before the line, lines 50-58 and 48-51, respectively, to install the required user-written packages)

- Run dofile “Dofiles/_masterscript.do” to produce the entire analysis.

Tables and programs

The provided code reproduces:

- ☒ All numbers provided in text in the paper
- ☒ All tables and figures in the paper

Figure/Table #	Program	Line Number	Output file	Note
Table 2	Dofiles/AEJ_Tables_3 Jun 2025.do	971	opportunity_costs.doc	
Table B.1	Dofiles/AEJ_Tables_3 Jun 2025.do	113	balance_test_results_child.doc	
Table B.2	Dofiles/AEJ_Tables_3 Jun 2025.do	143	balance_test_results_household.doc	
Table B.3	Dofiles/AEJ_Tables_3 Jun 2025.do	182	total_schooling.doc	
Table B.4	Dofiles/AEJ_Tables_3 Jun 2025.do	207	total_schooling_clustering.doc	
Table B.5	Dofiles/AEJ_Tables_3 Jun 2025.do	260, 602, 701, 729, 861	total_schooling_smaller_MOB_windows_mics.doc; occupation_smaller_MOB_windows.doc; childbearing_smaller_MOB_windows.doc; marriage_smaller_MOB_windows.doc; hiv_smaller_MOB_windows.doc	
Table B.6	Dofiles/AEJ_Tables_3 Jun 2025.do	401	total_schooling_HH_fixed_effects.doc	
Table B.7	Dofiles/AEJ_Tables_3 Jun 2025.do	1004	migration.doc	
Table B.8	Dofiles/AEJ_Tables_3 Jun 2025.do	449	skills_adults.doc	
Table B.9	Dofiles/AEJ_Tables_3 Jun 2025.do	488	economic.doc	
Table B.10	Dofiles/AEJ_Tables_3 Jun 2025.do	534	HH_wealth.doc	

Table B.11	Dofiles/AEJ_Tables_3 Jun 2025.do	638	childbearing.doc
Table B.12	Dofiles/AEJ_Tables_3 Jun 2025.do	673	marriage.doc
Table B.13	Dofiles/AEJ_Tables_3 Jun 2025.do	833	hiv_by_gender.doc
Table B.14	Dofiles/AEJ_Tables_3 Jun 2025.do	798	childbearing_and_marriage_by_age.doc
Table B.15	Dofiles/AEJ_Tables_3 Jun 2025.do	899	intergenerational.doc
Table B.16	Dofiles/AEJ_Tables_3 Jun 2025.do	308, 342	total_schooling_livestock_ages_5-10.doc; total_schooling_livestock_ages_11-18.doc
Table B.17	Dofiles/AEJ_Tables_3 Jun 2025.do	1040	decomposition.doc
Figure 1	Dofiles/AEJ_Figures_3 Jun 2025.do	116	MOB_and_SchoolEntry.jpg
Figure 2	Dofiles/AEJ_Figures_3 Jun 2025.do	201	Density_tests.jpg
Figure 3	Dofiles/AEJ_Figures_3 Jun 2025.do	219	placebo_maternal_education.jpg
Figure 4	Dofiles/AEJ_Figures_3 Jun 2025.do	295	Years_of_education.jpg
Figure 5	Dofiles/AEJ_Figures_3 Jun 2025.do	334	Reversal_of_fortunes.jpg
Figure 6	Dofiles/AEJ_Figures_3 Jun 2025.do	474	School_attainment.jpg
Figure 7	Dofiles/AEJ_Figures_3 Jun 2025.do	508	literacy_ICT_skills_adults.jpg
Figure 8	Dofiles/AEJ_Figures_3 Jun 2025.do	543	economic_outcomes.jpg
Figure 9	Dofiles/AEJ_Figures_3 Jun 2025.do	631	health_outcomes.jpg
Figure 10	Dofiles/AEJ_Figures_3 Jun 2025.do	879	intergenerational.jpg
Figure 11	Dofiles/AEJ_Figures_3 Jun 2025.do	324	years_of_education_by_gender_rddplot.jpg
Figure 12	Dofiles/AEJ_Figures_3 Jun 2025.do	355	Reversal_of_fortunes_gender.jpg

Figure 13	Dofiles/AEJ_Figures_ 596	economic_outcomes_g ender.jpg
Figure 14	Dofiles/AEJ_Figures_ 684	health_outcomes_gend er.jpg
Figure 15	Dofiles/AEJ_Figures_ 684	HIV_gender.jpg
Figure 16	Dofiles/AEJ_Figures_ 721	family_formation_age.j pg
Figure 17	Dofiles/AEJ_Figures_ 812	timeinnursery.jpg
Figure 18	Dofiles/AEJ_Figures_ 757	skills_kids.jpg
Figure B.1	Dofiles/AEJ_Figures_ 156	MOB_and_SchoolEntry _census.jpg
Figure B.2	Dofiles/AEJ_Figures_ 136	AverageAge.jpg
Figure B.3	Dofiles/AEJ_Figures_ 935	Basutoland.png
Figure B.4	Dofiles/AEJ_Figures_ 281	placebo_childcharateri stics.jpg
Figure B.5	Dofiles/AEJ_Figures_ 229	placebo_adultheight.jp g
Figure B.6	Dofiles/AEJ_Figures_ 963	effect sizes.png
Figure B.7	Dofiles/AEJ_Figures_ 390	Probability_of_attendi ng_gradeX_or_higher.j pg
Figure B.8	Dofiles/AEJ_Figures_ 411	School_leaving_by_age _grade.jpg
Figure B.9	Dofiles/AEJ_Figures_ 846	actual_cohort_educ.jpg
Figure B.10	Dofiles/AEJ_Figures_ 800	readingscore_sacmeq.j pg

References

Arnold, G. and B. Depew (2018). School starting age and long-run health in the united states. *Health Economics* 27(12), 1904–1920.

Basutoland (1965). *Annual Report of the Basutoland Director of Education for the year 1965*, Maseru, Basutoland, 1965.

- Black, S. E., P. J. Devereux, and K. G. Salvanes (2011). Too young to leave the nest? The effects of school starting age. *The Review of Economics and Statistics* 93(2), 455–467.
- Bureau of Statistics. 2019. *Lesotho Multiple Indicator Cluster Survey 2018, Survey Findings Report*. Maseru, Lesotho: Bureau of Statistics
- Fredriksson, P., K. Huttunen, and B. Öckert (2022). School starting age, maternal age at birth, and child outcomes. *Journal of Health Economics* 84, 102637.
- Fredriksson, P. and B. Öckert (2014). Life-cycle effects of age at school start. *The Economic Journal* 124(579), 977–1004.
- Guo, C., X. Wang, and C. Meng (2023). Does the early bird catch the worm? evidence and interpretation on the long-term impact of school entry age in China. *China Economic Review* 77, 101900.
- ICF. 2004-2017. Demographic and Health Surveys (various) [Datasets]. Funded by USAID. Rockville, Maryland: ICF [Distributor].
- Kawaguchi, D. (2011). Actual age at school entry, educational outcomes, and earnings. *Journal of the Japanese and International Economies* 25(2), 64–80.
- McCrary, J. and H. Royer (2011). The effect of female education on fertility and infant health: evidence from school entry policies using exact date of birth. *American Economic Review* 101(1), 158–195.
- Minnesota Population Center (2019). Integrated Public Use Microdata Series (IPUMS), International: Version 7.2 [Dataset]. IPUMS: Minneapolis, <https://www.ipums.org/projects/ipums-international/d020.v7.2>.
- Southern and Eastern Africa Consortium for Monitoring Educational Quality (SACMEQ). SACMEQ II Project 2000-2002 [dataset]. Harare: SACMEQ [producer], 2005. Paris: International Institute for Educational Planning (IIEP), UNESCO [distributor], 2000-2002. Accessed [10 April 2013] from UNESCO IIEP [via DISC]
- Southern and Eastern Africa Consortium for Monitoring Educational Quality (SACMEQ). SACMEQ III Project 2007 [dataset]. Version 1. Harare: SACMEQ [producer], 2007. Paris: International Institute for Educational Planning (IIEP), UNESCO [distributor], 2007. Accessed [10 April 2013] from UNESCO IIEP [via DISC]
- Valdés, M. T. and M. Requena (2024). The effect of the age at school entry on educational attainment and field of study: an analysis using the Spanish census. *Higher Education* 87(4), 1061–1083.
- World Bank (2024). World Development Indicators, GDP per capita (current US\$) [Data file]. <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD>. Retrieved December 7, 2024.